Tux Burner

User guide, command-line reference and glossary · version 0.99

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Tux Burner is a fork of PenguinBurner by jpietek (https://github.com/jpietek/PenguinBurner), GPL-3.0-or-later. It is an independent fork, not endorsed by or affiliated with the author of PenguinBurner.
Tested on one PC only (Ryzen 7 7800X3D + Radeon RX 9070 XT): other GPUs may have bugs.

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Tux Burner - User Guide

Version 0.99. Package `tux-burner`. Programs: `tux-burner` (also `tburn`), `tux-burner-cli` (also `tburn-cli`), and the root service `tux-burnerd`.

Tux Burner lets you run your graphics card on **less power with the same speed** (undervolting), or a little faster (overclocking), and saves the settings as profiles. It works on **AMD Radeon** cards through the amdgpu driver (tested on a Radeon RX 9070 XT) and on NVIDIA cards through the original PenguinBurner code.

Credit. Tux Burner is Copyright (C) 2026 by Eugene Flanagin. Licensed under GPL-3.0-or-later.

Tux Burner is a fork of **PenguinBurner** by **jpietek** (https://github.com/jpietek/PenguinBurner), GPL-3.0-or-later. The windows, the Auto-UV search, the adaptive per-game runtime, the Game Library, the overlay and the daemon's profile engine are the upstream author's work. This fork adds the AMD backend and renames the program. All upstream licence notices are kept.

Not affiliated. Tux Burner is an independent fork. It is not endorsed by or affiliated with the author of PenguinBurner. Please report Tux Burner problems to https://github.com/eugenesflanagin-spec/penguin-burner-amd/issues, not to upstream.

Tested on one PC. Tux Burner's AMD side was tested only on a Ryzen 7 7800X3D with a Radeon RX 9070 XT. Other AMD generations, and the NVIDIA backend in this fork, may have bugs. If you find one, please open an issue with your hardware and the report.

Other pages:

- CLI-REFERENCE.md - every command-line option.
- GLOSSARY.md - every technical word, in plain English.
- ../amd.md - the AMD design notes (what maps to what in sysfs).

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1. What it is, in plain words

A graphics card is given more voltage than most chips need. **Undervolting** lowers the voltage a little. The card then uses less power, runs cooler and quieter, and (if you did not go too far) is just as fast. Go too far and the card becomes **unstable**: the picture glitches, the driver resets, or the whole PC freezes.

Tux Burner does three jobs:

1. **Shows** what your GPU is doing (voltage offset, clocks, power limit, temperatures, fan).
2. **Changes** those settings, but only when you press a button and confirm, and always inside limits it checks first.
3. **Searches** for a good setting for you (the *Auto-UV ladder*): it lowers the voltage one small step at a time, runs a stress test at each step, and remembers the deepest step that passed.

On an AMD card there is **one** voltage knob (a global offset in millivolts), one core-clock offset, a memory-clock limit, a power cap and a fan curve. On NVIDIA there is a whole voltage/frequency curve, which is why the NVIDIA search is more complicated.

Who it is for

- Owners of an AMD Radeon (RDNA 3 / RDNA 4 with the amdgpu driver) who want a quieter or cooler card.
- People who already use PenguinBurner on NVIDIA and want to try the fork.

What it is not

- It is not a benchmark. It is not a replacement for **LACT**; the two can live together (see LACT and Tux Burner).
- It is not tested on your card. Every GPU is different. Start small.

---

2. Read this first: safety

- **Nothing is written to the GPU until you press a button and confirm.** The window opens in a read-only state. *Preview writes (dry run)* shows what would be written and writes nothing.
- A **bad undervolt can freeze the PC** or force a reset. Save your work first. Test on a PC you can reboot.
- **Sustained load.** *Stress test* and the *Auto-UV ladder* run a heavy GPU load for minutes. Do not run them unattended, and watch the temperatures.

- Tux Burner has two **safety floors** for the voltage offset: how deep it is allowed to go. The default is **-150 mV**. The suggested settings it offers never go deeper than **-110 mV**. See The safety floors.
- If LACT is running, **LACT owns the GPU.** Tux Burner will refuse to write unless you say `--override-lact`.
- You run hardware tuning at your own risk. See also `SECURITY.md`.

---

3. Install

Arch Linux, CachyOS and other Arch-based systems

From a release (see the Releases page of the repository):

```
sudo pacman -U tux-burner-0.99-1-x86_64.pkg.tar.zst
```

The package `provides` and `conflicts` with `penguin-burner` and `penguin-burner-amd` and `replaces` `penguin-burner-amd`, so it takes over from either. Build it yourself from the source tree:

```
git clone https://github.com/eugenesflanagin-spec/penguin-burner-amd
cd penguin-burner-amd
git checkout v0.99          # or the branch you want
packaging/build-arch-local.sh
sudo pacman -U /path/shown/at/the/end/tux-burner-0.99-1-x86_64.pkg.tar.zst
```

Debian, Ubuntu, Fedora, openSUSE

Each release also has a `.deb` and a `.rpm` file, a SHA256SUMS file and this guide as a PDF. These are built by the release workflow; they have been tried less than the Arch package, so please report problems.

```
sudo apt install ./tux-burner_0.99-1_amd64.deb     # Debian / Ubuntu
sudo dnf install ./tux-burner-0.99-1.x86_64.rpm    # Fedora
```

Check the download first: `sha256sum -c SHA256SUMS --ignore-missing`.

What you need besides the package

- The **amdgpu** kernel driver with OverDrive enabled (on most distributions the kernel option `amdgpu.ppfeaturemask=0xffffffff` is needed to see `pp_od_clk_voltage`). Without it Tux Burner can show values but cannot change them.
- For the stress test: an **OpenCL** driver (`rocm-opencl-runtime`, or Mesa's rusticl).
- Optional: **LACT**, **RADV** (`vulkan-radeon`).

The root service (once)

Changing GPU settings needs root. Tux Burner does that with one small root service, `tux-burnerd`. Install it once, as your normal user:

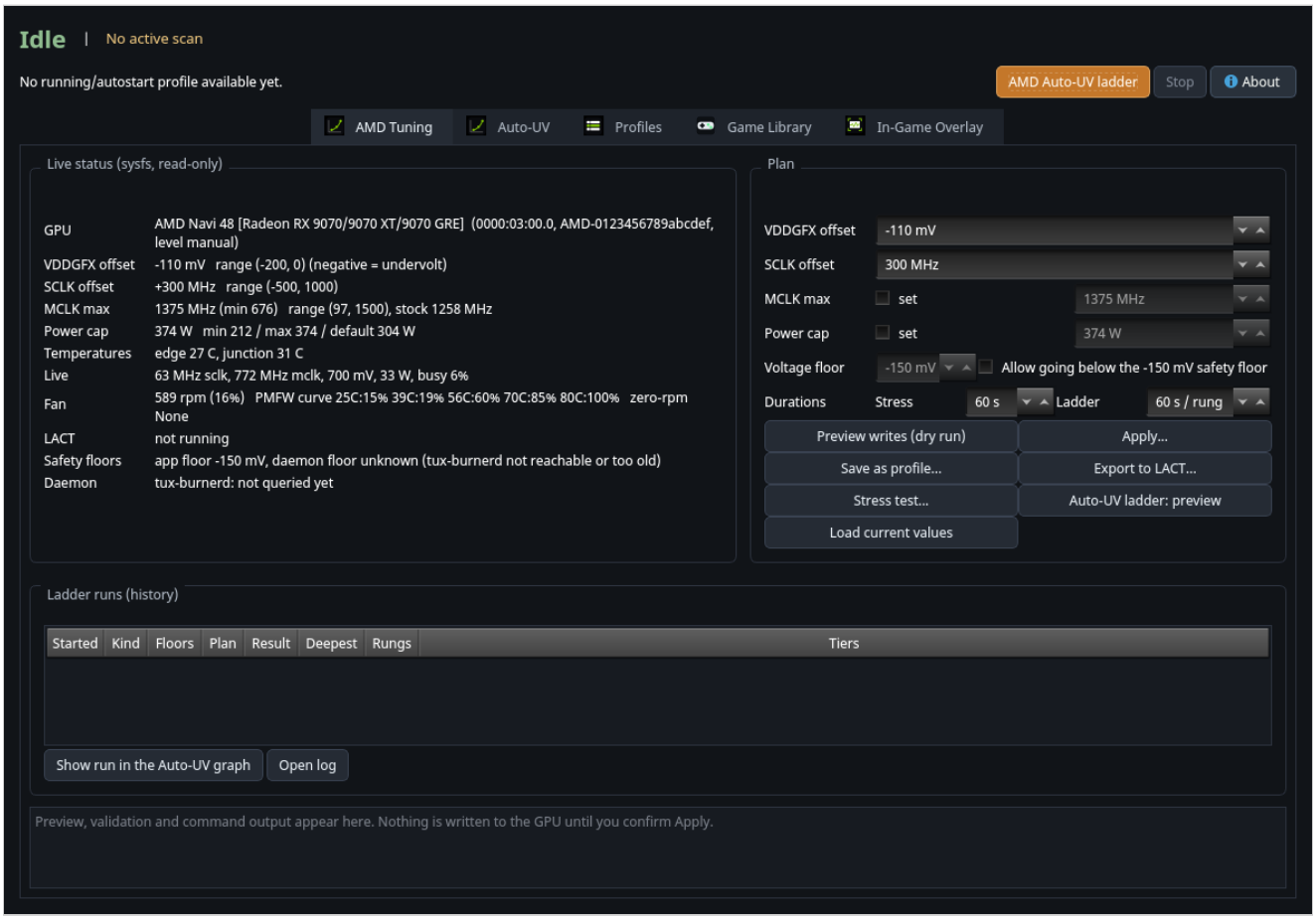
```
tburn-cli --install-systemd-service
```

It asks for your password (pkexec or sudo), installs and starts `tux-burnerd.service`. From then on the window and the command line talk to it over a local socket with no more password prompts. A fresh service applies **nothing** on start: it waits for you.

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4. First run

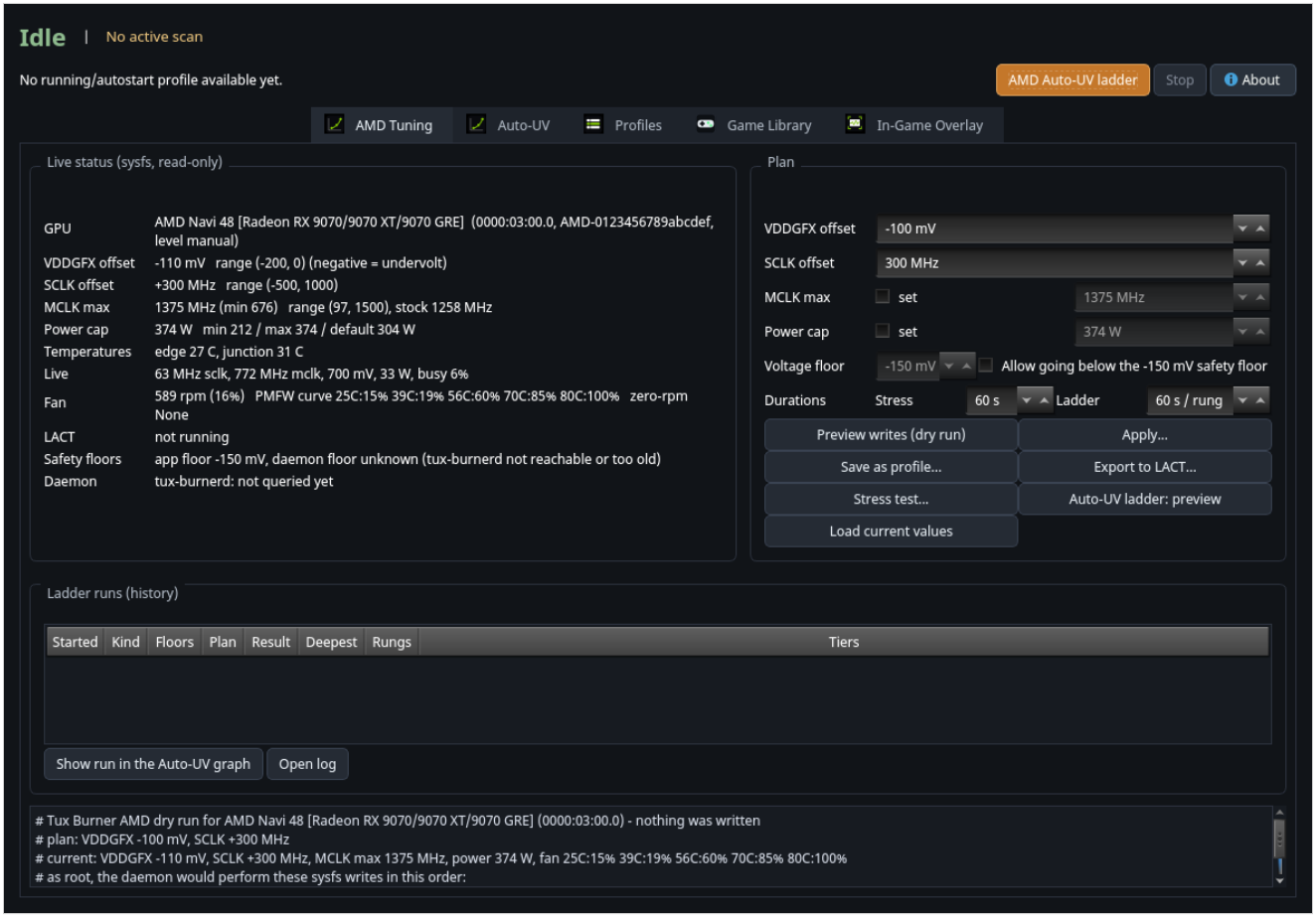
1. Start **Tux Burner** from the menu (or run `tux-burner`).
2. The window opens on the **AMD Tuning** tab. On the left, *Live status* shows your card: its name, the current VDDGFX offset, SCLK offset, MCLK max, power cap, temperatures, clocks, fan, whether LACT is running, and the two safety floors.



1. In a terminal, check the same thing without the window:

```
tburn-cli --amd-status
```

1. Try a **dry run**. On the right, change *VDDGFX offset* to a small value such as -50 mV and press **Preview writes (dry run)**. The text at the bottom lists exactly what would be written, and says "nothing was written".



1. Only when you are happy: **Apply...**, read the confirmation, and confirm.

If the page says "daemon floor unknown (tux-burnerd not reachable)", the root service is not running. Install it as shown above.

---

5. The window, page by page

The top bar shows the state (**Idle** / running), a one-line message, the big **AMD Auto-UV ladder** button, **Stop**, and **About**. The tabs are **AMD Tuning**, **Auto-UV**, **Profiles**, **Game Library** and **In-Game Overlay**.

5.1 AMD Tuning

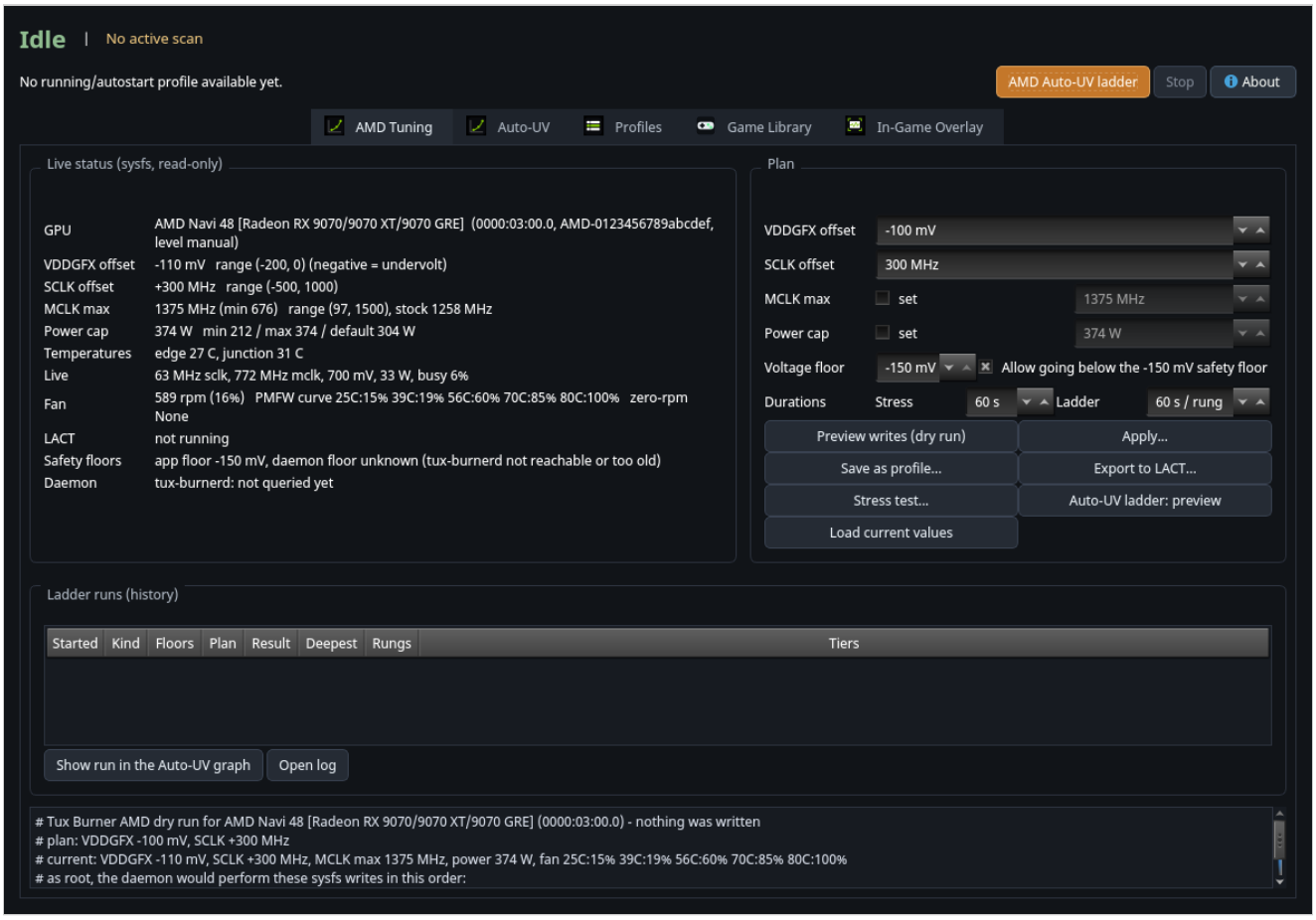
Live status (read from sysfs every 2 seconds, read-only):

Row	Meaning
GPU	Name, PCI address and an id derived from the card.

Row	Meaning
VDDGFX offset	The voltage offset in mV now, and the range the driver allows. Negative means undervolt.
SCLK offset	Core clock offset in MHz, and its range.
MCLK max	The memory clock ceiling, and the stock value for comparison.
Power cap	The power limit in watts, with the minimum, maximum and default the card reports.
Temperatures	Edge and junction (hot spot). The fan curve uses the junction.
Live	Core clock, memory clock, voltage, power and busy percent right now.
Fan	Fan speed and the card's 5-point fan curve.
LACT	Whether LACT is running and what it has configured.
Safety floors	The app floor and the daemon floor (they must match).
Daemon	What `tux-burnerd` reports.

Plan (what you want to set):

Control	What it does
VDDGFX offset	Voltage offset in mV. Cannot go below the safety floor.
SCLK offset	Core clock offset in MHz.
MCLK max (set)	Tick to set an absolute memory clock ceiling.
Power cap (set)	Tick to set the power limit in watts.
Voltage floor / Allow going below	Unlocks the floor spin box. A warning appears. See safety floors.
Durations	Seconds for the Stress test, and seconds per rung for the ladder.



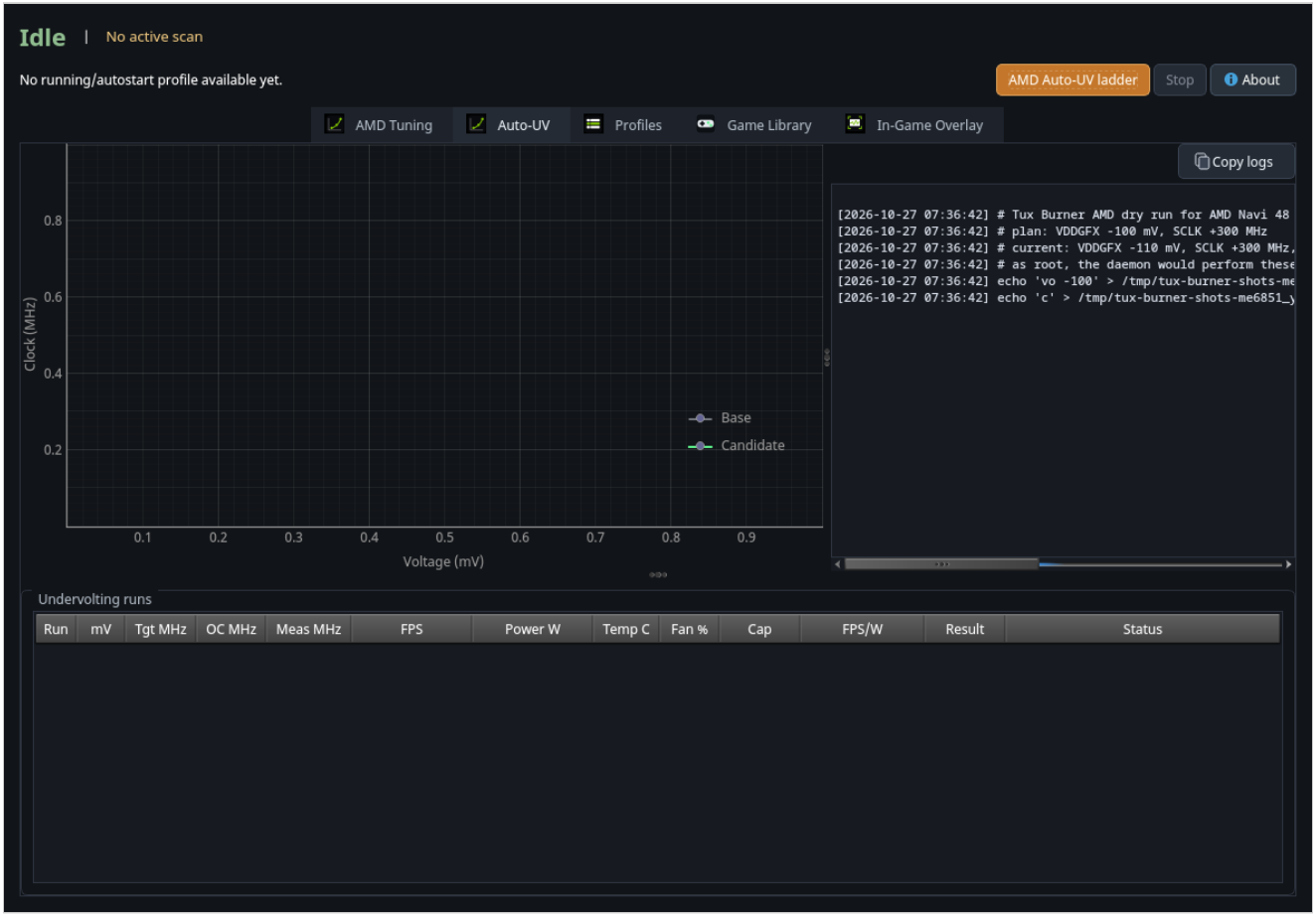
Buttons:

Button	What it does
Preview writes (dry run)	Shows the exact writes. Writes nothing.
Apply…	After a confirmation, writes the plan through the root service. With LACT running you first choose *Export to LACT*, *Override LACT* or *Cancel*.
Save as profile…	Saves the plan as a Tux Burner profile (see Profiles).
Export to LACT…	Writes the plan as a LACT configuration block you can paste into LACT's config.
Stress test…	Runs the OpenCL stress workload for the chosen seconds, after a confirmation. Heavy load.
Auto-UV ladder: preview	The first click previews the rungs; the second asks for confirmation and runs.
Load current values	Puts the card's present values back into the editor.

Ladder runs (history): every ladder run (from the window or the command line) is listed with its settings, floors, result and deepest stable rung. *Show run in the Auto-UV graph* plots it; *Open log* opens the log.

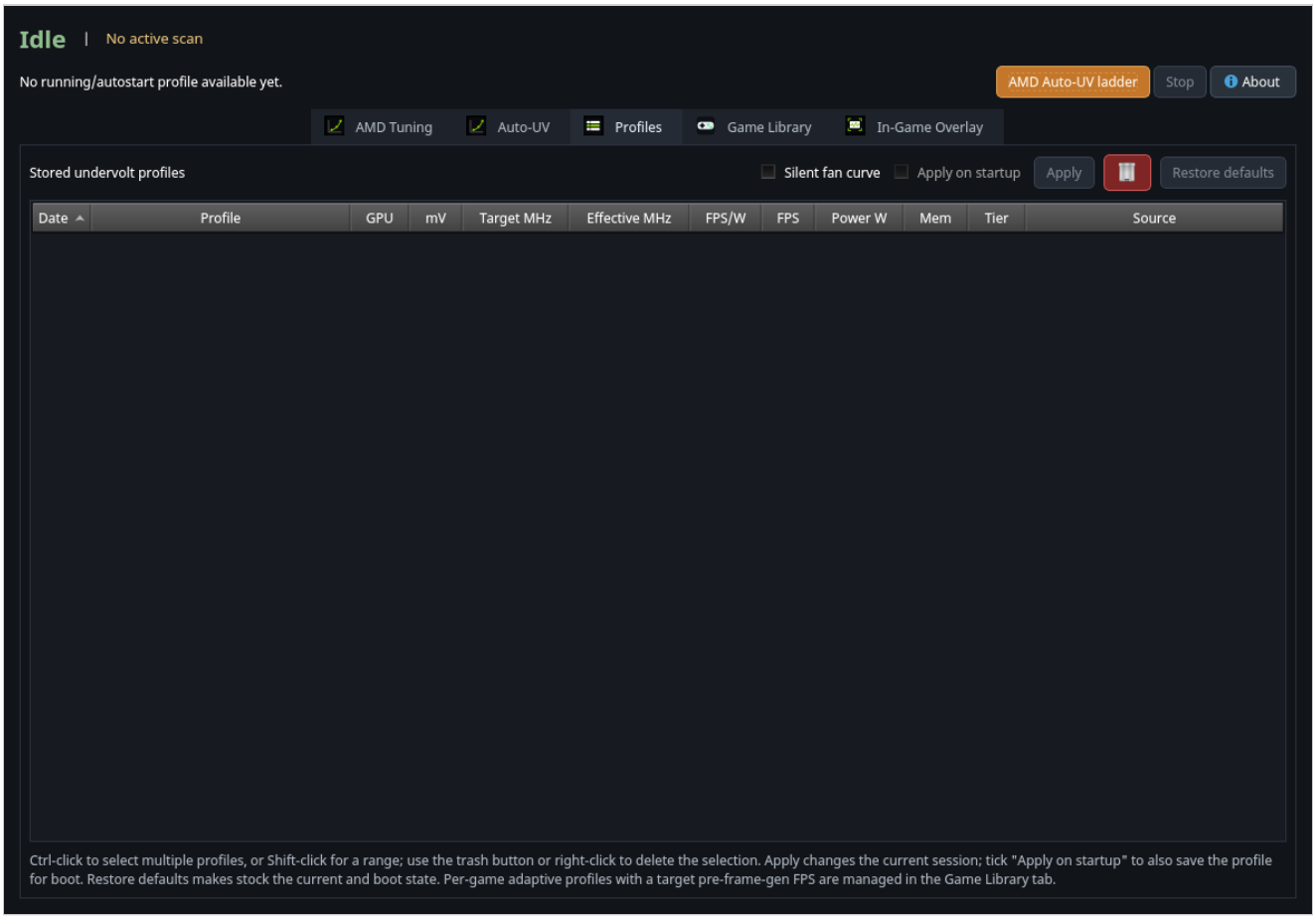
5.2 Auto-UV

The upstream NVIDIA Auto-UV page, with the live Clock-vs-Voltage graph. On AMD the graph shows each ladder rung's operating point, coloured by result. The *Start* button is replaced by *AMD Auto-UV ladder*.



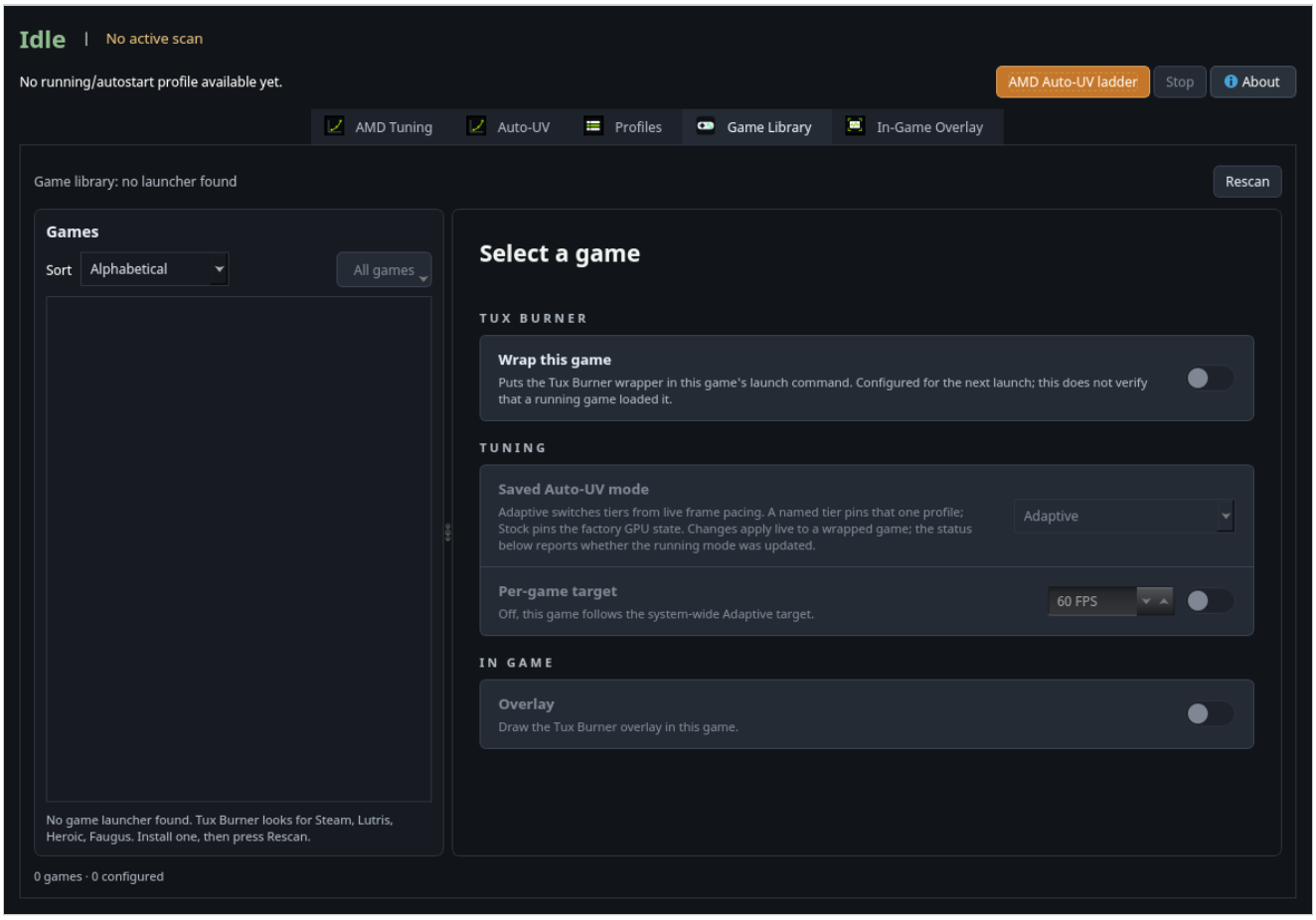
5.3 Profiles

Saved profiles, including AMD ones: name, tier, the AMD columns (voltage offset, core offset, memory, power), apply, rename, export to LACT, delete, and **Restore Defaults**. On AMD, Restore Defaults offers *Return to LACT's settings* (asks `lactd` to re-apply its profile; no restart) or *Driver stock defaults* (through the root service, with before/after readback).



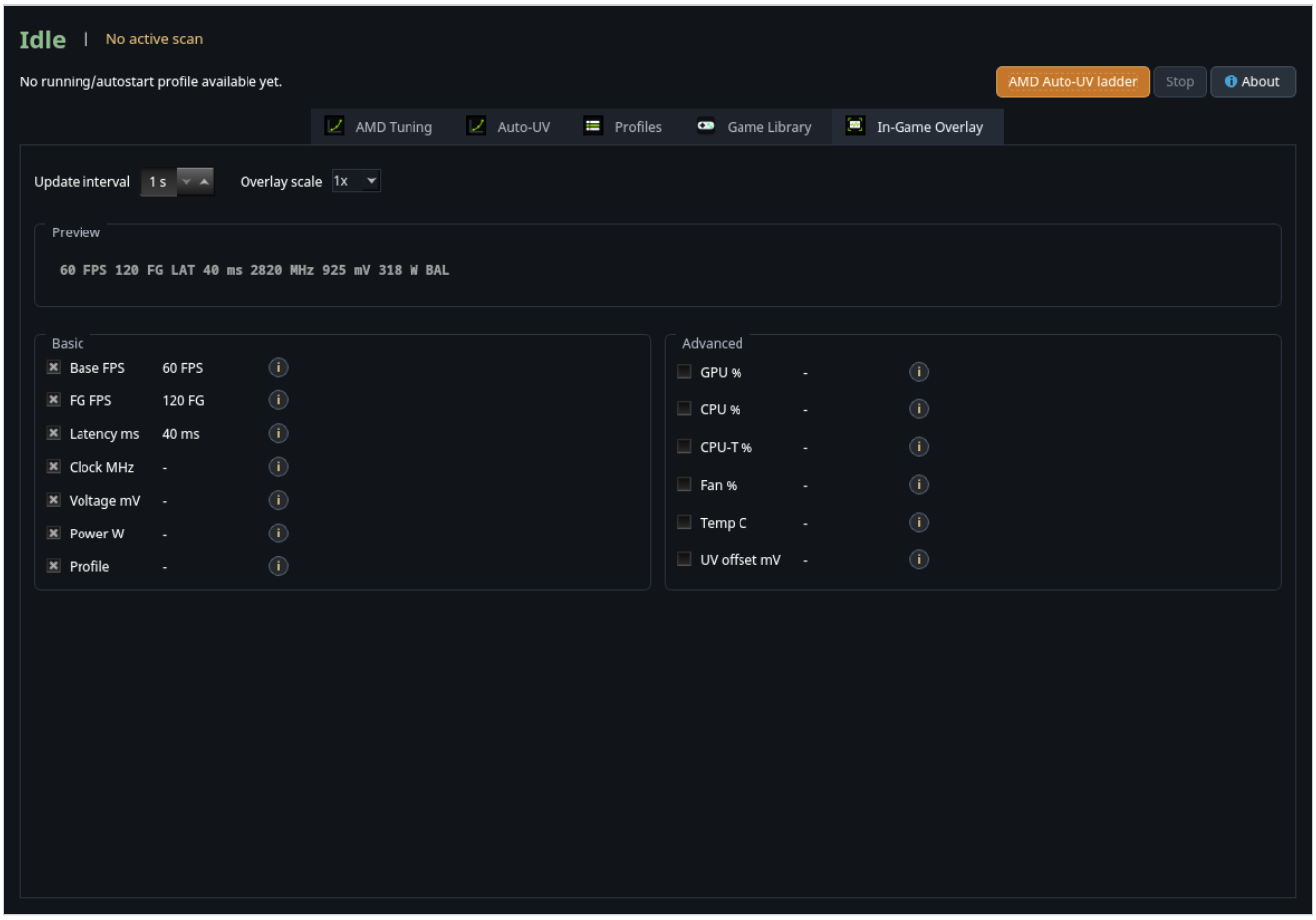
5.4 Game Library

The upstream per-game feature: Steam, Lutris, Heroic and Faugus games, each with its own tuning and overlay switch. Per-game profile switching was written for NVIDIA; on AMD only saved AMD profiles can be applied, and this part has had less testing.



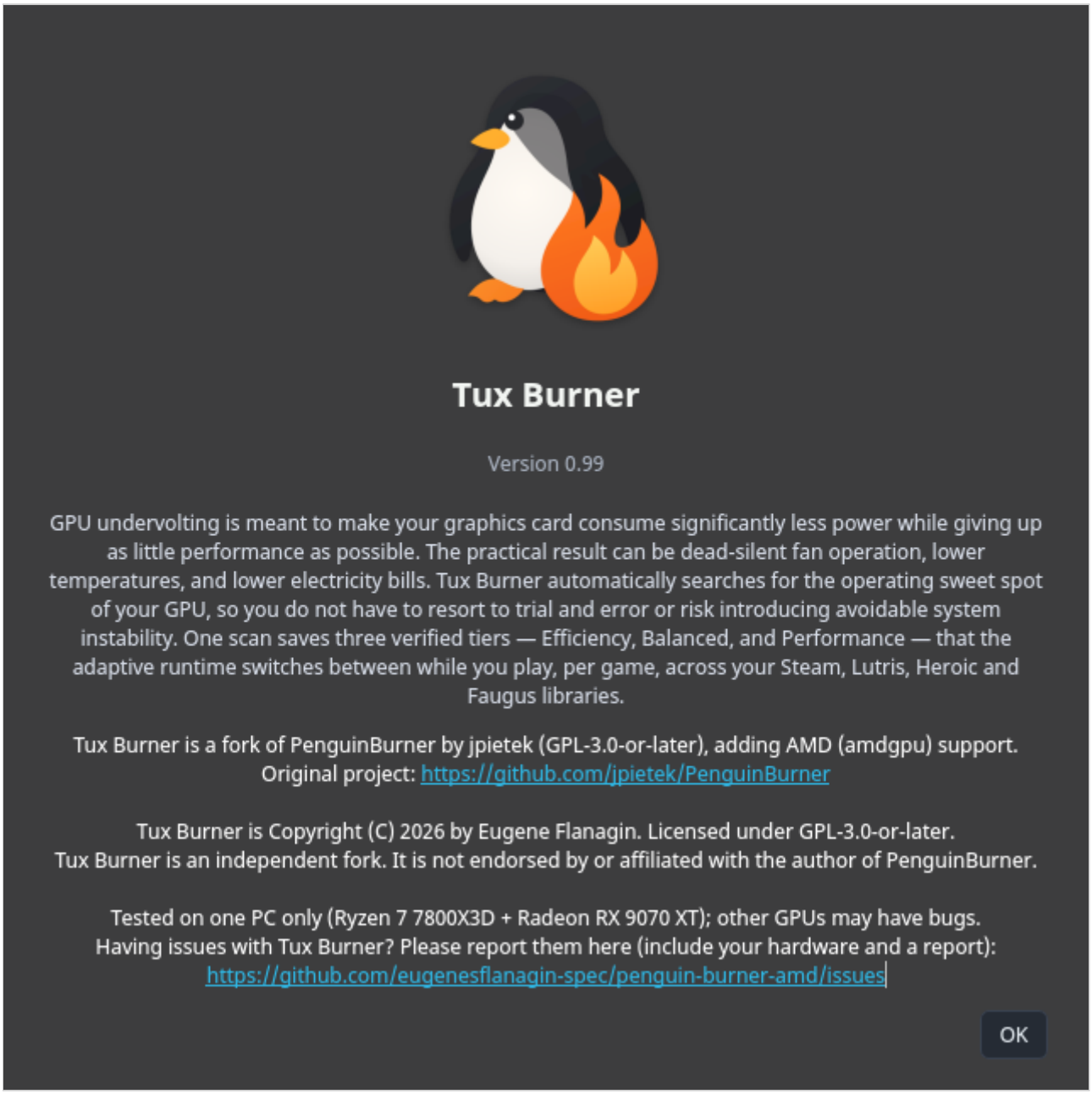
5.5 In-Game Overlay

The upstream Vulkan overlay layer (frame rate, GPU numbers, optional latency). It is independent of the AMD tuning code.



5.6 About

About shows the version, the fork notice, the upstream link and where to report problems.



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6. The safety floors

A **floor** is the deepest VDDGFX offset that will be allowed. There are two, on purpose, so that a bug in one cannot cause a deep undervolt:

Floor	Where it lives	Default
App floor	The window and `tburn-cli` (plan checks, the ladder).	-150 mV
Daemon floor	The root service. It refuses any deeper write no matter who asks.	-150 mV

How they behave:

- The default is **-150 mV**. The card itself may allow -200 mV; Tux Burner stops earlier.

- **Suggested** settings (the three tiers a ladder saves, presets, restore points) never go deeper than **-110 mV**, even though the ladder may *test* down to the floor.
- To go deeper on purpose you must do two things: raise the app floor (`--amd-voltage-floor MV --allow-below-floor`, or tick *Allow going below the safety floor* in the window) **and** lower the daemon floor (`tburn-cli --amd-set-daemon-floor MV --allow-below-floor`, or confirm the window's dialog, which does it for you). Until both match, *Apply* and the ladder are blocked in the window, so no rung is refused halfway and mistaken for a failure.
- The daemon's floor lives in `/etc/tux-burner/amd-vddgfx-floor-mv`. The environment variable `TUX_BURNERD_AMD_VDDGFX_FLOOR_MV` in the service's unit pins it instead (then the file is ignored and the window says so).
- Untick *Allow going below* (or run `--amd-set-daemon-floor default`) to return to -150 mV.

The floors limit the voltage offset only. The SCLK offset, memory clock and power cap are always checked against the ranges the driver reports.

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7. LACT and Tux Burner

LACT is a well-known Linux control panel for AMD GPUs. If its service (`lactd`) is running, **LACT owns the GPU's tuning and its fan curve**, and it re-applies its own settings from time to time.

What Tux Burner does about it:

- It detects LACT (the service is active, or its socket exists and a `lact` process runs) and shows the result and LACT's configured values in the window.
- `--amd-apply` and `--amd-auto-uv` **refuse** while LACT runs, unless you add `--override-lact`. In the window, *Apply* then offers *Export to LACT* (the preferred way), *Override LACT* (with a warning) or *Cancel*.
- It never touches the fan curve while LACT runs.
- `--amd-export-lact` writes your plan as a LACT `gpus:` block (voltage_offset, gpu_clock_offsets, max_memory_clock, power_cap and a curve of up to five points) so LACT can carry the setting. Put it in LACT's config yourself.
- **Never restart `lactd` on a live desktop to pick up a change.** Use LACT's own profile switching (`lact cli profile set <name>`) or reboot.

With `--override-lact`, LACT's own settings come back the next time LACT applies them. Restore LACT's values with *Return to LACT's settings* (Profiles > Restore Defaults) or `tburn-cli --amd-restore lact`.

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8. The Auto-UV ladder

On NVIDIA, Auto-UV searches a whole curve. On AMD there is a single voltage offset, so the search is a **ladder**:

1. Start at 0 mV (stock) and go down in steps of 10 mV (`--amd-auto-uv-step`).

2. At each **rung**, apply the offset and run the OpenCL stress workload for 60 seconds (`--amd-auto-uv-rung-seconds`). The workload checks its own results; a wrong answer, a driver error, a hang or a crash fails the rung.
3. Stop at the first failure or at the floor.
4. The deepest rung that passed, **plus a 20 mV margin** (`--amd-auto-uv-margin`), becomes three saved profiles: **Efficiency** (stock power cap), **Balanced** (the cap that was in force when the scan started) and **Performance** (your core offset and the maximum cap). Suggested voltages are never deeper than -110 mV.
5. A marker file is written before each rung. If the PC hard-resets during a rung, the next run reads the marker and treats that offset as **unsafe**.

Use *Auto-UV ladder: preview* (or `--amd-auto-uv --amd-dry-run`) first: it prints the planned range, for example `+0 -> -150 mV`, the number of rungs and the worst-case time. The ladder is heavy sustained load.

In the window, a live **Clock-vs-Voltage** plot shows each rung's measured operating point (voltage from hwmon against core clock), coloured by result, with the current rung highlighted. Each run is saved as `~/.config/TuxBurner/amd-runs/amd-run-*.json`.

Failure types shown in the history: OpenCL mismatch, driver error, hang timeout, process crash, setup error, and "apply refused" (by the floor, the driver range or LACT).

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9. Profiles

A profile is a saved plan. AMD profiles hold the VDDGFX offset, the core offset, the memory ceiling and the power cap.

- **Save:** *Save as profile...* in the window, or `tburn-cli --amd-plan vo=-90 sclk=0 power=304 --amd-save-profile quiet --amd-profile-tier efficiency`.
- **Tiers:** Efficiency, Balanced, Performance. The adaptive runtime (upstream feature) can switch between the tiers while you play; AMD profiles carry no per-point curve, so the drift check that could fight LACT does not apply.
- **Apply:** from the Profiles page. A profile that was saved by hand is marked *unverified*: it has not passed a ladder.
- **Boot profile:** the upstream *Apply on startup* workflow persists a profile in the root service. Leave it off if LACT applies your settings at boot, otherwise the two race.
- **Stock:** *Restore Defaults* (see above) or `tburn-cli --restore-stock`.

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10. Command line

`tburn-cli` and `tux-burner-cli` are the same program. The full list is in CLI-REFERENCE.md, `tburn-cli --help` and `man tburn-cli`. The AMD part, in short:

```
tburn-cli --version
tburn-cli --amd-status                                          # read-only
```

```
tburn-cli --amd-plan vo=-90 sclk=300 mclk=1375 power=374      # preview only
tburn-cli --amd-plan vo=-90 --amd-apply --amd-dry-run         # preview of an apply
tburn-cli --amd-plan vo=-90 --amd-apply                       # write (refused if
LACT runs)
tburn-cli --amd-plan vo=-90 --amd-apply --override-lact
# write although LACT runs
tburn-cli --amd-plan vo=-90 --amd-export-lact ~/lact-amd.yaml
tburn-cli --amd-stress 60                                     # OpenCL load; no
tuning writes
tburn-cli --amd-auto-uv --amd-dry-run                         # ladder plan
tburn-cli --amd-auto-uv                                       # run the ladder
(heavy load)
tburn-cli --amd-restore lact                                  # or: stock
tburn-cli --daemon-status
```

`power=` takes watts or a percent of the default cap (`power=110%`). `fan=` takes up to five `TEMP:PERCENT` points. `zero_rpm=on|off` is also accepted.

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11. Files and settings

Path	What
`~/.config/TuxBurner/`	Your profiles, ladder run history (`amd-runs/`), debug logs, settings (`tux_burner.toml`).
`/etc/tux-burner/amd-vddgfx-floor-mv`	The daemon's safety floor (root-owned).
`/var/lib/tux-burner/`	The daemon's saved state (boot profile, energy counters).
`/var/opt/tux-burner/libexec/tux-burnerd`	The installed copy of the daemon binary.
`/run/tux-burnerd.sock`	The daemon socket.
`/etc/systemd/system/tux-burnerd.service`	The service unit.

Environment variables: `TUX_BURNER_AMD_VDDGFX_FLOOR_MV` (app floor), `TUX_BURNERD_AMD_VDDGFX_FLOOR_MV` (daemon floor, in the unit), `TUX_BURNER_HOME` (use another home directory).

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12. Moving from PenguinBurner or penguin-burner-amd

Your data is **copied, never moved or deleted**:

- On install, the package copies `/etc/penguin-burner` to `/etc/tux-burner` and `/var/lib/penguin-burner` to `/var/lib/tux-burner` if the new ones do not exist yet.
- On first start, the program copies `~/.config/PenguinBurner` to `~/.config/TuxBurner` (profiles, run history, settings; `penguin_burner.toml` becomes `tux_burner.toml`).
- The daemon also reads the old `/etc/penguin-burner/amd-vddgfx-floor-mv` if the new floor file does not exist.

The **old service** (`penguin-burnerd.service`) is not touched by the package. Switch over yourself, once, so two daemons never run together:

```
sudo pacman -U tux-burner-0.99-1-x86_64.pkg.tar.zst
sudo systemctl disable --now penguin-burnerd.service
sudo rm -f /etc/systemd/system/penguin-burnerd.service
sudo tburn-cli --install-systemd-service
tburn-cli --daemon-status
tburn-cli --amd-status
```

Check that `--amd-status` shows your safety floor (and, if you had lowered it, the same value). Once everything looks right you may remove the old directories (`/etc/penguin-burner`, `/var/lib/penguin-burner`, `/var/opt/penguin-burner`, `~/.config/PenguinBurner`).

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13. Troubleshooting

Problem	What to try
"daemon floor unknown (tux-burnerd not reachable)"	Install the service: `tburn-cli --install-systemd-service`. Check `systemctl status tux-burnerd`.
The values cannot be changed	OverDrive is off. Check that `/sys/class/drm/card*/device/pp_od_clk_voltage` exists. Many distributions need `amdgpu.ppfeaturemask=0xffffffff` on the kernel command line.
A write is refused because LACT is running	LACT owns the GPU. Export to LACT, or use `--override-lact` knowing LACT will re-apply its values.
Apply or the ladder is greyed out	The app floor and the daemon floor differ. Use the window's *Safety floors* row to sync them.
"apply refused by the floor"	The value is deeper than the floor. Raise the floor on purpose (both floors), or pick a shallower value.
Stress test says no OpenCL device	Install an OpenCL driver (`rocm-opencl-runtime` or Mesa rusticl) and `ocl-icd`.

Problem	What to try
The screen freezes during a ladder	That offset was too deep. After the reset, run the ladder again: the marker file makes Tux Burner treat that offset as unsafe. Go back to LACT's values or a shallower offset.
After switching from PenguinBurner nothing is listed	See Moving from PenguinBurner. Check `~/.config/TuxBurner/`.
The Game Library or overlay does nothing	Those parts are upstream NVIDIA-era features and are less tested on AMD. Please report what you see.

When you report a problem, include `tburn-cli --version`, `tburn-cli --amd-status` and `tburn-cli --daemon-status`. **Remove** your host name, IP addresses and the GPU serial/unique id before posting.

---

14. Uninstall

```
sudo tburn-cli --uninstall-systemd-service      # stops and removes the service
sudo pacman -R tux-burner                       # or apt remove / dnf remove
```

Your files in `~/.config/TuxBurner/` are kept; delete them if you want. The GPU keeps whatever was last applied until the next reboot or until you restore it (`tburn-cli --amd-restore stock`).

---

15. Credits

- **PenguinBurner** by **jpietek**, GPL-3.0-or-later: https://github.com/jpietek/PenguinBurner. Based on version 0.8.3.
- **LACT** (https://github.com/ilya-zlobintsev/LACT) for the configuration format and the coexistence rules.
- The Linux **amdgpu** driver's OverDrive interface.
- See `CREDITS.md` and `CHANGELOG.md`.

Tux Burner is a fork of PenguinBurner. Changes are Copyright (C) 2026 by Eugene Flanagin. Licensed under GPL-3.0-or-later.

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16. Glossary

See GLOSSARY.md. The short version:

- **Undervolt**: lower the voltage so the card uses less power.
- **VDDGFX offset**: the AMD voltage knob, in millivolts; negative is lower.
- **SCLK / MCLK**: core clock / memory clock.

- **Rung**: one step of the ladder.
- **Floor**: the deepest offset allowed.
- **LACT**: a Linux AMD GPU control program.
- **Daemon**: the root service, `tux-burnerd`.

Tux Burner - Command-line reference

Version 0.99. Programs: `tburn-cli` and `tux-burner-cli` (the same program). `tux-burner` and `tburn` start the window.

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Plain words first: the commands that **only read** are `--version`, `--help`, `--amd-status`, `--daemon-status`, `--list-auto-uv-profiles`, and any `--amd-plan ...` **without** `--amd-apply` (that one prints what would be written). The commands that **write to the GPU** are `--amd-apply`, `--amd-auto-uv`, `--restore-stock`, `--amd-restore stock`, `--daemonize` and the upstream Auto-UV scan. `--amd-stress` and `--amd-auto-uv` also put heavy sustained load on the GPU. Write commands go through the root service `tux-burnerd`, and are refused while LACT runs unless you add `--override-lact`.

The AMD options in short

Option	What it does
`--amd-status`	Show ranges, live values, fan curve and LACT state. Read-only.
`--amd-plan KEY=VALUE ...`	Describe a plan: `vo=<mV>` `sclk=<MHz>` `mclk=<MHz>` `power=<W>` or `<NN%>` `fan=T:P,T:P,...` (up to 5 points) `zero_rpm=on
`--amd-dry-run`	With `--amd-apply`, `--amd-save-profile`, `--amd-auto-uv` or `--amd-stress`: show, change nothing.
`--amd-apply`	Write the plan now (checked against the live driver ranges and both floors).
`--amd-save-profile NAME`	Save the plan as a Tux Burner profile (marked unverified).
`--amd-profile-tier T`	Tier for the saved profile: `efficiency`, `balanced`, `performance`.
`--amd-export-lact [PATH]`	Write the plan as a LACT config fragment (to the screen without PATH).
`--amd-stress SECONDS`	OpenCL stress workload. Heavy load. No tuning writes.
`--amd-auto-uv`	The ladder. Heavy load. Options: `--amd-auto-uv-start MV` (0), `--amd-auto-uv-step MV` (10), `--amd-auto-uv-floor MV` (driver minimum, never below the app floor), `--amd-auto-uv-rung-seconds S` (60), `--amd-auto-uv-margin MV` (20).
`--amd-voltage-floor MV`	The app's floor (default -150). Deeper needs `--allow-below-floor`.
	Confirms a floor deeper than -150 mV.

Option	What it does
`--allow-below-floor`	
`--amd-set-daemon-floor MV|default`	Set the root service's own floor.
`--amd-restore lact|stock`	Give the GPU back to LACT, or reset to driver defaults.
`--override-lact`	Write although LACT is running.

Full `--help` output

This is the output of `tburn-cli --help`, kept here so the guide matches the program. The upstream sections (Auto-UV, Steam overlay setup, Runtime and daemon, Runtime tuning, Advanced/debug) describe the NVIDIA-era features; many of them are not used on AMD.

```
usage: tburn-cli [options]

Tux Burner: GPU undervolting and overclocking for Linux (AMD amdgpu
and NVIDIA). A fork of PenguinBurner by jpietek (GPL-3.0-or-later).
Run Auto-UV scans, manage the saved Efficiency/Balanced/Performance
profiles, drive the adaptive daemon runtime, and set up the Steam
launch wrapper and in-game overlay.

options:
  -h, --help            show this help message and exit
  --version             Print the Tux Burner version and exit

Auto-UV:
  --auto-uv-voltage-scan
                        Discover a stable fixed-clock undervolt from the
                        live/default NVIDIA V/F curve, step the lock voltage
                        down through real editable VF bins, and verify
                        candidates with Q2RTX plus CUDA load. Resume an
                        unfinished scan automatically when the GPU and
                        settings match
  --list-auto-uv-profiles
                        List saved Auto-UV profiles and exit; use a shown
                        profile id or candidate id with --auto-uv-profile.
  --assign-auto-uv-tier PROFILE TIER
                        Assign a saved verified Auto-UV profile to an adaptive
                        tier and exit. PROFILE accepts the same selectors as
                        --auto-uv-profile; TIER is efficiency, balanced,
                        performance, or none.
  --auto-uv-mode MODE   Auto-UV preset path. efficiency uses one 2-bin tail
                        sweep to the voltage floor; balanced uses the 2-bin
                        tail sweep; performance uses the 2-bin tail sweep plus
                        Auto-OC.
```

```
--auto-uv-q2rtx-resolution {auto,1080p,1440p,4k}
                      Q2RTX resolution for the complete Auto-UV scan and
                      final verification. Default: auto (4K on desktop RTX
                      4080/Super, 4090, 5070 Ti, 5080 and 5090, plus RTX
                      4090/5090 Laptop; 1080p on all others, including
                      unknown GPUs). Use 1080p to compare a lighter
                      rendering workload. CUDA is unchanged.
--auto-uv-retry-after-freeze
                      CLI only; off by default. On an explicit scan resume
                      after a hard freeze, allow retrying the interrupted
                      clock climb at higher voltage. May require further
                      reboots. Keeps the failed-point blacklist, predicted
                      stability margins, floor caution and voltage headroom.
--auto-uv-min-voltage-mv mV
                      Lowest voltage for the selected undervolt point.
                      Overrides the detected GPU table floor and the
                      percentage-drop fallback. Lower-voltage curve
                      transitions are checked separately at their lower
                      clocks.
--auto-uv-memory-offset-mhz MHz
                      Memory clock V/F offset to apply during Auto-UV and
                      save with the final profile. NVML offsets are in
                      transfer-rate units (MT/s): the realized memory clock
                      rises by half the value. Clamped to the driver-
                      reported limit for the GPU (fallback 0..2000 when NVML
                      exposes no range).
--auto-uv-power-limit-w W
                      Power limit in watts to apply during Auto-UV and save
                      with the final profile. The UI clamps this to the
                      selected GPU's NVML minimum and maximum power-limit
                      range.
--auto-uv-boost {standard,aggressive}
                      CLI-only boost headroom for every scanned tier:
                      Standard Boost (default) uses 2 rising tail bins (+30
                      MHz nominal); Aggressive Boost uses 4 (+60 MHz
                      nominal). Does not enable freeze retries.
--auto-uv-tail-rise-bins N
                      How many voltage bins can the voltage curve rise above
                      the locked undervolt point. Defaults to 2 for every
                      tier; 0 makes the tail flat.
--auto-uv-final-verification-s SECONDS
                      Final verification soak per tier for this scan (every
                      stage of an adaptive 3-in-1 run uses the same
                      duration). Default 300 seconds; equivalent to
                      TUX_BURNER_AUTO_UV_FINAL_SECONDS.
--auto-uv-efficiency-power-limit-w W
                      Full-scan override: the Efficiency tier's power limit
                      in watts, applied for that tier's baseline, descent,
                      final verification, and saved profile. Only meaningful
                      with --auto-uv-mode adaptive.
--auto-uv-efficiency-memory-offset-mhz MHz
```

```
                      Full-scan override: the Efficiency tier's memory V/F
                      offset (NVML transfer-rate units, like --auto-uv-
                      memory-offset-mhz), applied for that tier's descent
                      and saved with its profile. Only meaningful with
                      --auto-uv-mode adaptive.
  --auto-uv-efficiency-target-voltage-mv mV
                      Custom Efficiency voltage target for a single-tier or
                      full scan. Defaults use the GPU table. Clock ranges:
                      Efficiency -15% to Balanced; Balanced from Efficiency
                      to Performance; Performance from Balanced to
                      Performance +5%. Change only if you understand GPU
                      tuning.
  --auto-uv-efficiency-target-clock-mhz MHz
                      Custom Efficiency clock target for a single-tier or
                      full scan. Defaults use the GPU table. Clock ranges:
                      Efficiency -15% to Balanced; Balanced from Efficiency
                      to Performance; Performance from Balanced to
                      Performance +5%. Change only if you understand GPU
                      tuning.
  --auto-uv-balanced-power-limit-w W
                      Full-scan override: the Balanced tier's power limit in
                      watts, applied for that tier's baseline, descent,
                      final verification, and saved profile. Only meaningful
                      with --auto-uv-mode adaptive.
  --auto-uv-balanced-memory-offset-mhz MHz
                      Full-scan override: the Balanced tier's memory V/F
                      offset (NVML transfer-rate units, like --auto-uv-
                      memory-offset-mhz), applied for that tier's descent
                      and saved with its profile. Only meaningful with
                      --auto-uv-mode adaptive.
  --auto-uv-balanced-target-voltage-mv mV
                      Custom Balanced voltage target for a single-tier or
                      full scan. Defaults use the GPU table. Clock ranges:
                      Efficiency -15% to Balanced; Balanced from Efficiency
                      to Performance; Performance from Balanced to
                      Performance +5%. Change only if you understand GPU
                      tuning.
  --auto-uv-balanced-target-clock-mhz MHz
                      Custom Balanced clock target for a single-tier or full
                      scan. Defaults use the GPU table. Clock ranges:
                      Efficiency -15% to Balanced; Balanced from Efficiency
                      to Performance; Performance from Balanced to
                      Performance +5%. Change only if you understand GPU
                      tuning.
  --auto-uv-performance-power-limit-w W
                      Full-scan override: the Performance tier's power limit
                      in watts, applied for that tier's baseline, descent,
                      final verification, and saved profile. Only meaningful
                      with --auto-uv-mode adaptive.
  --auto-uv-performance-memory-offset-mhz MHz
                      Full-scan override: the Performance tier's memory V/F
```

```
                        offset (NVML transfer-rate units, like --auto-uv-
                        memory-offset-mhz), applied for that tier's descent
                        and saved with its profile. Only meaningful with
                        --auto-uv-mode adaptive.
  --auto-uv-performance-target-voltage-mv mV
                        Custom Performance voltage target for a single-tier or
                        full scan. Defaults use the GPU table. Clock ranges:
                        Efficiency -15% to Balanced; Balanced from Efficiency
                        to Performance; Performance from Balanced to
                        Performance +5%. Change only if you understand GPU
                        tuning.
  --auto-uv-performance-target-clock-mhz MHz
                        Custom Performance clock target for a single-tier or
                        full scan. Defaults use the GPU table. Clock ranges:
                        Efficiency -15% to Balanced; Balanced from Efficiency
                        to Performance; Performance from Balanced to
                        Performance +5%. Change only if you understand GPU
                        tuning.
  --auto-oc-target-voltage-mv mV
                        Performance-mode Auto-OC voltage cap in mV. Overrides
                        the detected GPU table target. Defaults are optimized
                        for most GPUs; change only if you understand
                        voltage/frequency tuning and instability risks.
  --auto-oc-target-clock-mhz MHz
                        Performance-mode Auto-OC core clock cap in MHz.
                        Overrides the detected GPU table target. Defaults are
                        optimized for most GPUs; change only if you understand
                        voltage/frequency tuning and instability risks.

Steam overlay setup:
  --set-steam-overlay-launch APPID
                        Write the Tux Burner native-overlay launch option for
                        a Steam app id and exit. The write is verified so
                        running Steam rewrites are reported instead of treated
                        as success.

Runtime and daemon essentials:
  --silent-fan-curve    Runtime/daemon only: opt in to Tux Burner manual fan-
                        curve control; by default fan control is left to the
                        GPU driver. Auto-UV scans write a suggested fan curve
                        automatically when safe.
  --daemonize           Apply normal runtime through the running tux-burnerd
                        service after an Auto-UV final curve exists. Auto-UV
                        scans remain foreground-only.
  --install-systemd-service
                        Install or update the Tux Burner daemon service:
                        refresh the daemon binary at /var/opt/tux-
                        burner/libexec, rewrite the unit, and restart it. With
                        profile options, also set the persistent boot profile;
                        without them, an existing boot profile is kept. Asks
                        for authorization (pkexec/sudo) when not run as root.
```

```
  --uninstall-systemd-service, --deinstall-systemd-service
                        Stop and remove the persistent Tux Burner systemd
                        service. Asks for authorization (pkexec/sudo) when not
                        run as root.
  --migrate-to-daemon-service
                        Install tux-burnerd.service and migrate an existing
                        legacy TuxBurner.service when possible. Asks for
                        authorization (pkexec/sudo) when not run as root.
  --daemon-status       Print Tux Burner hardware daemon status and exit.
  --set-main-gpu GPU-UUID
                        On a multi-NVIDIA-GPU system, make a saved startup GPU
                        the daemon's main monitored GPU after boot.
  --clear-main-gpu      Clear the explicit Main GPU and restore last-saved-GPU
                        behavior.
  --restore-stock       Recovery: reset the GPU to stock now and make stock
                        the boot state through the running tux-burnerd
                        service. Saved profiles are kept. Works without the
                        GUI.
  --auto-uv-profile AUTO_UV_PROFILE
                        Use an Auto-UV profile by profile id, candidate id,
                        JSON path, 'active', or 'latest' for daemon runtime,
                        systemd service runtime, and internal final
                        verification.
  --adaptive-auto-uv    Runtime/daemon only: allow Tux Burner to adapt between
                        saved Auto-UV profile tiers from base present-frame
                        p95 pacing. Requires at least one saved verified
                        profile tier. Target defaults to 60 FPS; override the
                        service env TUX_BURNER_ADAPTIVE_TARGET_FPS for 30, 50,
                        60, 120, etc.
  --journal-hours N     Hours of systemd journal history to suggest after
                        daemonizing; default 4.

Runtime tuning:
  --config CONFIG       Runtime config path to read defaults from
  --gpu-index GPU_INDEX
                        Override the configured GPU index

Advanced/debug:
  --debug-log           Write a verbose diagnostic log next to the selected
                        config file under debug-logs/; with the default config
                        this is ~/.config/TuxBurner/debug-logs

AMD (amdgpu OverDrive) tuning:
  Local AMD port. Reads come from sysfs; writes go through tux-burnerd. While LACT
(lactd) runs it owns the GPU: writes need --override-lact.

  --amd-status          Print the AMD GPU's OverDrive ranges, live values, fan
                        curve and LACT state (read-only)
  --amd-plan KEY=VALUE [KEY=VALUE ...]
                        Tuning plan: vo=<mV> (VDDGFX offset, negative =
                        undervolt), sclk=<MHz> offset, mclk=<MHz> max memory
```

```
                      clock, power=<W>|<NN%> cap, fan=T:P,T:P,... (max 5
                      points), zero_rpm=on|off. Alone it previews the exact
                      sysfs writes (dry run)
  --amd-dry-run       With --amd-apply/--amd-save-profile/--amd-auto-
                      uv/--amd-stress: show what would happen, change
                      nothing
  --amd-apply         Apply --amd-plan now through tux-burnerd (validated
                      against the live OD_RANGE)
  --amd-save-profile NAME
                      Save --amd-plan as an (unverified) Tux Burner profile
                      the GUI and daemon can apply
  --amd-profile-tier {efficiency,balanced,performance}
                      Tier to assign the saved AMD profile to
  --amd-export-lact [PATH]
                      Write --amd-plan as a LACT config fragment (stdout
                      when PATH is omitted)
  --amd-stress SECONDS  Run the OpenCL brute-force stability workload
                      (sustained GPU load; no tuning writes)
  --amd-auto-uv       AMD Auto-UV: step the VDDGFX offset down rung by rung,
                      stress each rung, save three tiers
  --amd-auto-uv-start MV
                      First rung (default 0 = stock)
  --amd-auto-uv-step MV
                      Rung size in mV (default 10)
  --amd-auto-uv-floor MV
                      Lowest offset to try (default: driver minimum)
  --amd-auto-uv-rung-seconds S
                      Stress seconds per rung (default 60)
  --amd-auto-uv-margin MV
                      Safety margin added to the deepest stable rung
                      (default 20)
  --amd-voltage-floor MV
                      Deepest VDDGFX offset Tux Burner will plan, write or
                      sweep to (default -150 mV on this machine). Going
                      deeper also needs --allow-below-floor; the daemon
                      enforces its own floor (/etc/tux-burner/amd-vddgfx-
                      floor-mv)
  --allow-below-floor   Confirm an --amd-voltage-floor deeper than the default
                      -150 mV
  --amd-set-daemon-floor MV|default
                      Set the DAEMON's VDDGFX safety floor (/etc/tux-
                      burner/amd-vddgfx-floor-mv, written by tux-burnerd) or
                      'default' to restore -150 mV. Values deeper than -150
                      also need --allow-below-floor. Without this the daemon
                      refuses deeper writes even if --amd-voltage-floor
                      allows them
  --amd-restore {lact,stock}
                      Restore the AMD GPU: 'lact' asks lactd to re-apply its
                      current profile (lact cli profile set); 'stock' resets
                      OverDrive (sclk/mclk/voltage offsets), the power cap
                      and - unless LACT runs - the PMFW fan curve to driver
```

```
                        defaults through tux-burnerd, with readback
  --override-lact       Write even though LACT is running (LACT re-applies its
                        own settings on its next apply)
```

Exit status

`0` success. Non-zero: a refused or failed operation (the message says why). The OpenCL stress workload itself exits with 0 = passed, 1 = setup error (no device, build failed), 3 = instability (a wrong result), 4 = hang (the GPU stopped answering); the ladder reads these to decide a rung's result.

Tux Burner - Glossary

Every technical word in the guide, in plain English.

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Word	Plain meaning
Adaptive runtime	An upstream feature: while you play, the daemon switches between your Efficiency, Balanced and Performance profiles depending on the frame rate.
amdgpu	The Linux kernel driver for AMD Radeon cards. Tux Burner talks to it through files under `/sys/class/drm/`.
App floor	The deepest voltage offset the window and `tburn-cli` allow. Default -150 mV.
Auto-UV	"Automatic undervolt": the search for a good voltage setting. On AMD it is the *ladder*.
Balanced	A tier: the power cap that was set when the scan started, with the found voltage.
Boot profile	A profile the root service applies when the PC starts. Off by default.
Clock	How fast a part of the GPU runs, in MHz. SCLK is the core clock, MCLK the memory clock.
Daemon	A program that runs in the background. Here: `tux-burnerd`, the root service that does every GPU write.
Daemon floor	The root service's own limit on the voltage offset. It refuses deeper writes whatever the window says.
Dry run	Show what would be done and do nothing.
Edge / junction	Two GPU temperatures. Junction (the hot spot) is the hotter one; the fan curve follows it.
Efficiency	A tier: the found voltage with the card's default power cap.
Fan curve	A list of "at this temperature, run the fan at this percent" points. The AMD firmware accepts up to five.
Floor	See *App floor* and *Daemon floor*.
Fork	A copy of a project that goes its own way. Tux Burner is a fork of PenguinBurner.
GPL-3.0-or-later	The free-software licence of Tux Burner and of PenguinBurner.
hwmon	The kernel's folder of sensor readings (temperature, power, fan, voltage).
Instability	The card makes wrong results, glitches, resets the driver or freezes the PC.
LACT	"Linux AMDGPU Control Application", a popular tool for tuning AMD cards. Its service is `lactd`.
Ladder	The AMD Auto-UV search: 0 mV, -10, -20 ... one *rung* at a time, with a stress test on each.
Margin	Extra safety added to the deepest passing rung (default 20 mV, so the saved value is shallower).
MCLK max	The ceiling of the memory clock, in MHz.
mV	Millivolt, a thousandth of a volt.

Word	Plain meaning
OpenCL	A standard for running calculations on the GPU. Tux Burner's stress test uses it on AMD.
OverDrive	AMD's name for tuning controls in the driver. Tux Burner reads and writes them through `pp_od_clk_voltage`.
Overlay	The in-game frame-rate display (an upstream feature).
Performance	A tier: your core offset and the highest power cap.
PMFW	The firmware on the GPU that runs the fan curve.
Power cap	The most power, in watts, the card may use.
Profile	A saved plan (voltage, clocks, power, tier).
Rung	One step of the ladder.
SCLK offset	How many MHz to add to (or take from) the core clock.
Stock	The driver's default settings.
Stress test	A heavy GPU workload that checks its own answers, used to see whether a setting is stable.
sysfs	The kernel's folder of files under `/sys` that expose hardware.
Tier	One of Efficiency, Balanced, Performance.
Undervolt	Lower the voltage. Less power and heat, if the card stays stable.
VDDGFX offset	The AMD voltage knob: a global offset in mV added to the card's voltage table. Negative is an undervolt.
Zero RPM	The mode in which the fans stop at low temperature.